Necessary, but Also Preferred: Why Attention Sinks Form Even When Transformers Can Switch Off

Aditya Kariwal, Manish Kundu & Samyak Sheersh
Group G28 — Foundations of Large Language Models (End-Term Project)

Abstract

Ran-Milo (2026) recently proved that softmax attention must form an attention sink to implement a trigger-conditional "no-op", and suggested that explicit gating or non-normalised attention is the right way to obtain sink-free models. We test this suggestion in the paper's own controlled setting. We re-implement the task, reproduce its results, and add three attention mechanisms that have an explicit "off" switch: softmax-off-by-one, a query-dependent output gate, and sigmoid attention. We first construct explicit sink-free solutions for gated and off-by-one attention, showing that for them a sink is not necessary. We then show, over more than 300 training runs, that gradient-based training forms sinks anyway: gated attention puts ≥ 0.95 of its attention on the bos token at every depth from 1 to 8 layers, even when the gate is initialised almost closed; off-by-one attention only moves mass into its null slot as depth grows. Removing the bos token flips the picture: both mechanisms then learn to use their switch, while softmax fails. Gradient probes show that the loss pulls harder towards closing the gate than towards the sink, yet the sink wins. We explain this as a speed race: the sink is a product of two learnable factors ($q^\top k$) and grows self-reinforcingly, while the gate is linear in its weights. Consistent with this, raising only the gate's learning rate by $10\times$ removes all sinks in a 4-layer model and makes training $5\times$ faster.

1 Introduction

Large language models put a disproportionate share of attention on the first token, a phenomenon called the attention sink (Xiao et al., 2024). Sinks are linked to massive activations (Sun et al., 2024), complicate quantisation (Bondarenko et al., 2023), and matter for long-context inference. A common explanation is that a head sometimes has nothing useful to read; since softmax weights must sum to one, the head "parks" its attention on a token whose value contributes nothing (Miller, 2023; Gu et al., 2025; Guo et al., 2024).

Ran-Milo (2026) made this precise. On a trigger-conditional task, where a head must average the past only when a trigger token appears and output zero otherwise, they prove that any low-loss softmax model must put almost all attention on bos at non-trigger positions, while ReLU attention solves the same task without a sink. Their practical conclusion (Sec. 5) is that removing sinks requires changing how "off" states are represented, "via non-normalized attention, explicit gating", and they assume, citing Qiu et al. (2025), that sinks do not appear in gated attention.

This project asks whether that suggestion holds. If a model can switch a head off without a sink, does training actually use that ability? We study this in the exact setting of the original paper, where everything can be measured and controlled.

Contributions. Beyond reproducing the original results with our own implementation, we:

1. add three attention variants with an explicit off switch (softmax-off-by-one, gated softmax, sigmoid) and give explicit sink-free zero-loss constructions for gated and off-by-one attention (Section 5), so for them a sink is provably not necessary;
2. show that training nevertheless chooses the sink: across depths 1–8, gate initialisations from 0.02 to 0.98 and 5 seeds each, gated attention keeps full sinks and never uses its gate as a switch; off-by-one attention only drifts towards its null slot in deep models (Section 6);
3. show that without a bos token both mechanisms do learn to switch off, while softmax fails and forms no substitute anchor, contrary to a prediction in the original paper;
4. identify the mechanism (Section 7): gradient probes show the loss favours closing the gate early in training, yet the bilinear sink grows faster; and confirm it by intervention: a 10× larger learning rate on the gate parameters alone removes all sinks and speeds up training 5×.

2 Background and related work

Attention sinks. Xiao et al. (2024) showed that keeping the first tokens in the KV cache is essential for streaming inference. Gu et al. (2025) studied when sinks emerge during pretraining, finding that they appear after effective optimisation, are weaker with small learning rates, and disappear with un-normalised sigmoid attention. Sun et al. (2024) linked sinks to massive activations, and Guo et al. (2024) explained them with active–dormant heads. Varre et al. (2026) show that gradient flow on value×softmax models drives outputs to low-entropy solutions, a formal route to sinks.

Sinks are necessary for softmax. Ran-Milo (2026) prove that on trigger-conditional tasks softmax attention needs a sink (single-layer, Thm. 1; at least one layer of a deep model, Thm. 2), while ReLU attention admits a sink-free solution (Thm. 3). Their theorems extend to any attention whose weights are normalised ($\sum_k \alpha_{ik} \geq c > 0$) and monotone; the variants we add below violate exactly these conditions.

Mechanisms that can switch off. Softmax-off-by-one adds a constant 1 to the softmax denominator, letting rows sum to less than one (Miller, 2023). Bondarenko et al. (2023) proposed clipped softmax and gated attention to help heads "do nothing". Qiu et al. (2025) add a head-specific sigmoid gate after attention and report strongly reduced sinks in LLMs trained on 3.5T tokens. Sigmoid attention (Ramapuram et al., 2025) removes normalisation altogether.

Closest work. Two very recent preprints are directly relevant. Rizwan et al. (2026) report that gating did not remove sinks in small, short-trained language models, and speculate that the gate "never learned to be sparse", but do not test why. Wang (2026) find that a gate depending only on the value leaves the sink in place while a query-controlled mechanism removes it. Our study gives a controlled explanation of the first observation (a speed race, Section 7) and refines the second: our gate is query-dependent and still loses to the sink at default learning rates.

3 Setup

Task. We follow Ran-Milo (2026) and their released code exactly. A sequence has $L = 16$ tokens $x^{(0)}, \ldots, x^{(15)} \in \mathbb{R}^{16}$. Position 0 is bos (indicator e_{bos}, zero content). One position $j \in \{2, \ldots, 15\}$ is the trigger (e_{trig}); all others are content tokens (e_{cont}). Every non-bos token carries 13 i.i.d. $\mathcal{U}(-1, 1)$ content features. The target is the mean of $x^{(1)}, \ldots, x^{(j)}$ at the trigger and zero elsewhere (multi-layer models predict this on top of the residual stream). There is no positional encoding.

Models. Attention-only transformers with D layers of H heads, each head of width $d = 16$, causal masking, residual connections for $D > 1$, and no MLP or LayerNorm, as in the original paper. With $s_{ik} = q_i^\top k_k$, the head output is $W_O\big(g_i \odot \sum_{k \leq i} \alpha_{ik} v_k\big)$ with $g_i \equiv 1$

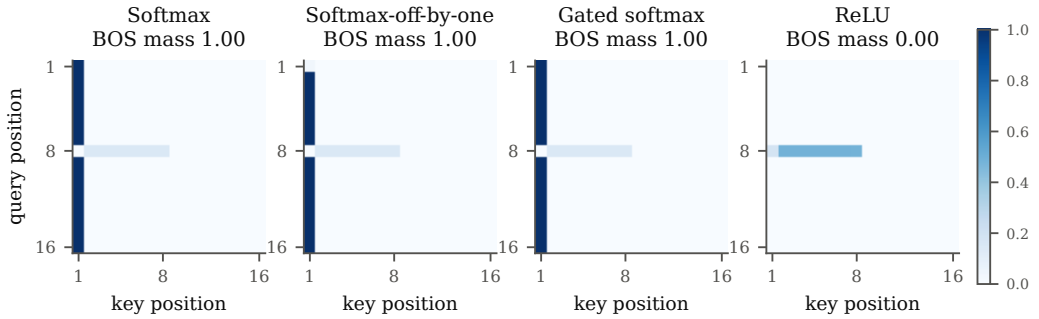


Figure 1: Mean attention of a trained single-layer, single-head model (trigger at position 8, 1000 test sequences, seed 0). Softmax, off-by-one and gated attention all park non-trigger rows on bos (column 1); ReLU switches off instead.

unless gated, and

$$\text{softmax:}\ \ \alpha_{ik} = \frac{e^{s_{ik}}}{\sum_{k' \le i} e^{s_{ik'}}}, \qquad \text{off-by-one:}\ \ \alpha_{ik} = \frac{e^{s_{ik}}}{1+\sum_{k' \le i} e^{s_{ik'}}},$$

$$\text{gated:}\ \ \text{softmax with } g_i = \sigma(W_G x_i + b_G), \qquad \text{ReLU:}\ \ \alpha_{ik} = \mathrm{ReLU}(s_{ik})/i,$$

$$\text{sigmoid:}\ \ \alpha_{ik} = \sigma(s_{ik} - \log L), \qquad \text{sigmoid-n:}\ \ \alpha_{ik} = \sigma(s_{ik})/i.$$

The gate is elementwise and head-specific and is computed from the query token, as in Qiu et al. (2025). "Sigmoid-n" uses the same $1/i$ length scaling as the paper's ReLU.

Training. Adam (Kingma & Ba, 2015) with $\beta = (0.9, 0.95)$, batch 128, learning rate 10^{-3} (1 layer, no residual) or 10^{-4} (residual models), weights $\mathcal{N}(0, 0.02^2)$, gate bias 0 unless stated. As in the reference code, training stops when the ℓ_∞ error on a batch falls below 5×10^{-3} (the paper's text says 10^{-2}) or after 50k steps (80k for 6–8 layers). Every configuration is run with 5 seeds.

Metrics. Following the paper, we evaluate on 1000 sequences with the trigger at position 8 and measure, per head, the attention mass on bos averaged over non-trigger rows ($i \ge 1$, $i \ne j$). A head is a sink head if this exceeds 0.6. For off-by-one we also report the null-slot mass $1 - \sum_k \alpha_{ik}$, and for gated attention the mean gate value on non-trigger rows and on the trigger row.

4 Reproduction

Figure 1 and Table 1 reproduce the paper: softmax forms a perfect sink in every seed and ReLU never does. All 90 runs converge except un-normalised sigmoid, which cannot compute an average at 1–2 layers and only succeeds once a 4-layer residual stack can compensate. Sigmoid-n solves the task without a sink at every depth, consistent with Gu et al. (2025). At 4 layers, sinks appear in some heads but not all (37/80 heads over 5 seeds), matching the existential form of the paper's Theorem 2.

5 A sink is not necessary when the head can switch off

The proofs of Ran-Milo (2026) rely on the head's output being a normalised average of values. Gating multiplies that average by a query-dependent factor, and off-by-one lets the weights sum to less than one, so neither is covered. We show both admit sink-free solutions, in the spirit of the paper's ReLU construction (Thm. 3). Let $W_K = W_V = W_O = I$ and write the score as $s_{ik} = x_i^\top B\, x_k$.

Proposition 1 (Gated attention). Let $B = -c\,(e_{\text{trig}} + e_{\text{cont}})e_{\text{bos}}^\top$ and $g_i = \sigma\big(a\,(x_i[\text{trig}] - x_i[\text{cont}] - x_i[\text{bos}])\big)\mathbf{1}$. Then every non-bos query gives score $-c$ to bos and 0 to all other keys, so $\alpha_{i,\text{bos}} = e^{-c}/(e^{-c} + i) \le e^{-c}$ at every row. The gate equals $\sigma(a)$ at the trigger

Table 1: Phase-1 results (5 seeds each). "Max BOS": largest per-head bos mass on non-trigger rows, mean ± std over seeds. "Sink heads": heads with bos mass > 0.6, summed over seeds. DLHH: D layers, H heads.

Model	Attention	Conv.	Steps	Max BOS	Sink heads	Gate off / trig.
1L1H	softmax	5/5	905	$1.00 \pm .00$	5/5	–
	off-by-one	5/5	911	$1.00 \pm .00$	5/5	–
	gated	5/5	1002	$1.00 \pm .00$	5/5	0.51 / 0.79
	ReLU	5/5	692	$0.00 \pm .00$	0/5	–
	sigmoid	0/5	–	–	–	–
	sigmoid-n	5/5	831	$0.18 \pm .03$	0/5	–
2L2H	softmax	5/5	10509	$1.00 \pm .00$	20/20	–
	off-by-one	5/5	8921	$1.00 \pm .00$	20/20	–
	gated	5/5	9794	$1.00 \pm .00$	20/20	0.45 / 0.66
	ReLU / sigmoid-n	5/5	–	0.00 / 0.13	0/20	–
4L4H	softmax	5/5	22379	$1.00 \pm .00$	37/80	–
	off-by-one	5/5	18155	$0.89 \pm .10$	40/80	–
	gated	5/5	16816	$0.99 \pm .01$	38/80	0.47 / 0.59
	ReLU / sigmoid-n	5/5	–	0.00 / 0.16	0/80	–

Table 2: Explicit sink-free solutions (Propositions 1 and 2) evaluated on 2000 sequences. The training stopping threshold is $\ell_\infty < 5 \times 10^{-3}$.

	Gated			Softmax-off-by-one			
Scale	MSE	ℓ_∞	BOS mass	MSE	ℓ_∞	BOS mass	Row mass
2	1.5×10^{-3}	0.174	2.9×10^{-2}	1.8×10^{-2}	0.625	6.6×10^{-2}	0.51
5	5.2×10^{-6}	0.010	1.5×10^{-3}	2.3×10^{-4}	0.086	6.4×10^{-3}	0.056
10	2.4×10^{-10}	10^{-4}	1.0×10^{-5}	1.2×10^{-8}	6×10^{-4}	4.5×10^{-5}	4×10^{-4}
20	≈ 0	≈ 0	4.6×10^{-10}	≈ 0	≈ 0	2.1×10^{-9}	2×10^{-8}

and $\sigma(-a)$ elsewhere. Since inputs are bounded, the maximum error over all inputs is $O(e^{-a} + e^{-c})$. Hence for any $\varepsilon > 0$ there are weights with loss $\le \varepsilon$ and bos mass $\le \varepsilon$ at all non-trigger rows.

Proposition 2 (Softmax-off-by-one). Let $B = m\big[e_{\text{trig}}(e_{\text{trig}} + e_{\text{cont}} - e_{\text{bos}})^\top - (e_{\text{bos}} + e_{\text{cont}})(e_{\text{bos}} + e_{\text{trig}} + e_{\text{cont}})^\top\big]$. The trigger query scores $+m$ on every non-bos key and $-m$ on bos, so its weights tend to the uniform average over positions $1..j$. Every other query scores $-m$ on all keys, so its total mass is $\le i\, e^{-m}$ and it outputs ≈ 0 through the null slot. Again the error is $O(Le^{-m})$ and the bos mass vanishes.

Table 2 evaluates both constructions numerically on 2000 random sequences. Already at scale 10, well within the range reached by trained weights, both reach an error far below the training threshold with essentially zero bos attention. Notably, the gated solution uses a query-independent attention pattern and puts all of the conditional behaviour into the gate.

6 … but training learns the sink anyway

Gated attention always sinks. Despite Proposition 1, every trained gated model in Table 1 parks non-trigger rows on bos. The gate barely moves from its initial value of 0.5: it ends at about 0.45–0.5 on non-trigger rows and 0.6–0.8 on the trigger row, a mild modulation rather than a switch. The sink does the switching.

Initialising the gate closed makes it worse. One might expect a gate that starts nearly closed to be used as a switch. Figure 2 shows the opposite. With the gate initialised at $\sigma(-4) = 0.02$ or $\sigma(-2) = 0.12$, all 80/80 heads become sinks (vs. 38/80 at 0.5), and in every

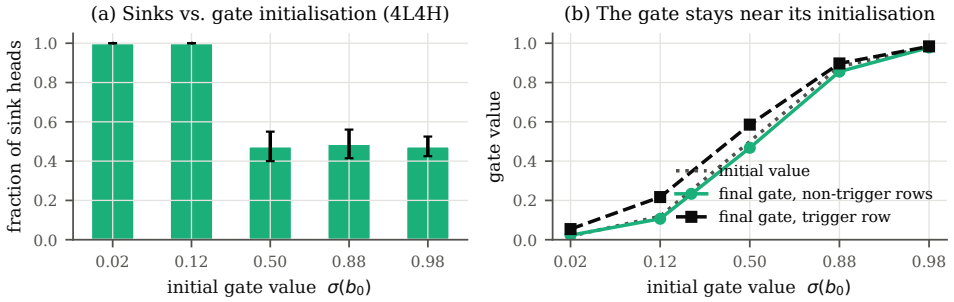


Figure 2: Gated attention, 4 layers × 4 heads, 5 seeds per bar. (a) A nearly closed initial gate turns every head into a sink. (b) The trained gate stays near its initialisation (dotted) instead of becoming an on/off switch.

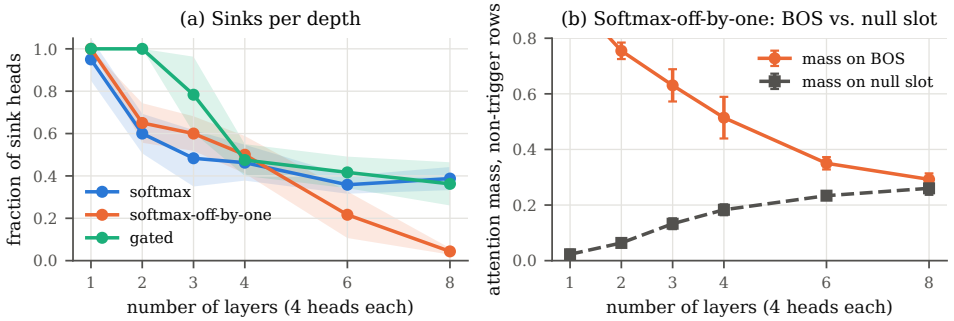


Figure 3: Depth sweep (4 heads per layer, 5 seeds; band/bars: ±1 std). (a) Fraction of sink heads. (b) For off-by-one, attention mass on bos vs. the null slot, averaged over all heads.

setting the final gate stays close to where it started (e.g. 0.02 → 0.02/0.05 off/trigger). The model rescales W_V, W_O around a frozen gate and still builds a sink.

Depth. Figure 3 sweeps depth from 1 to 8 layers with 4 heads per layer. Softmax and gated attention keep sinks at every depth (a declining fraction of heads is needed as depth grows, but sink heads always have bos mass ≥ 0.95). Off-by-one behaves differently: at 1–2 layers it is indistinguishable from softmax, but with depth its sinks fade (7/160 sink heads at 8 layers) and mass moves into the null slot. Within the heads that remain sinks, the null-slot share grows from 0.02 (1–2 layers) to 0.11, 0.19, 0.32 and 0.33 (3, 4, 6, 8 layers).

Removing the bos token. If training prefers the sink only because a convenient anchor exists, it should use the off switch once the anchor is gone. We drop bos: every position is an ordinary content token, and there is no positional information at all. Table 3 confirms the prediction. Softmax cannot solve the task at any depth (0/15 runs), off-by-one and gated attention always do, and the gated model now uses a genuine switch (gate 0.02 off vs. 0.62 on at one layer). Softmax also forms no substitute anchor (no head puts more than 0.2 on any single key). This contradicts the expectation of Ran-Milo (2026, Sec. 5) that suppressing the sink makes an anchor "reappear elsewhere". In our setting no other token can be singled out without positional information, so this result may not hold for models with positional encodings.

7 Why the sink wins: a speed race

Timelines. Figure 4 tracks every head during training. Sinks appear abruptly, with an S-shaped rise between roughly steps 2k and 5k, and never reverse in softmax and gated

Table 3: Training without a bos token (5 seeds per cell). "Mass": mean total attention $\sum_k \alpha_{ik}$ on non-trigger rows (1 for softmax and gated by construction).

Attention	1 layer, 1 head	2 layers, 2 heads	4 layers, 4 heads
softmax	0/5 ($\ell_\infty = 0.74$)	0/5 (0.70)	0/5 (0.08)
off-by-one	5/5, mass 0.00	5/5, mass 0.42	5/5, mass 0.57
gated	5/5, gate 0.02 / 0.62	5/5, gate 0.28 / 0.63	5/5, gate 0.41 / 0.64
ReLU, sigmoid-n	5/5	5/5	5/5

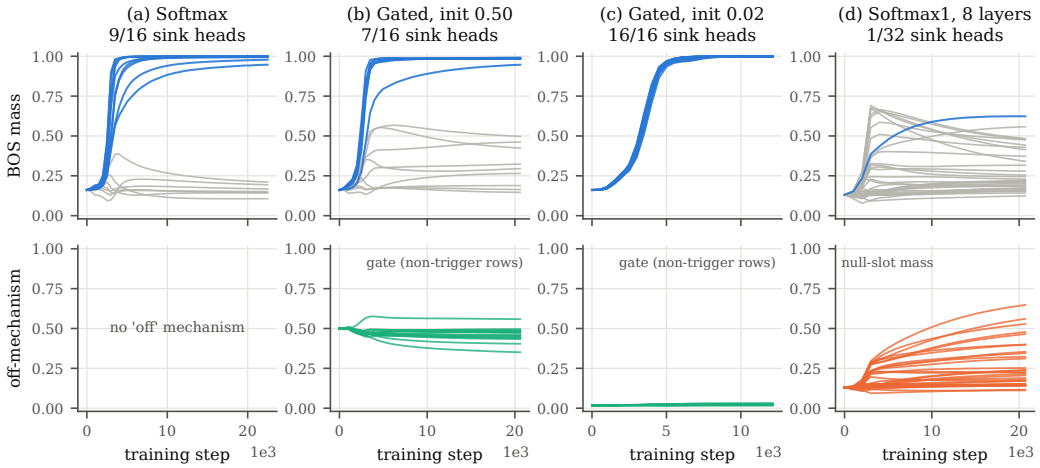


Figure 4: Training timelines (seed 0, 4 layers × 4 heads unless noted). Top: bos mass per head (blue: heads that end as sinks). Bottom: the head's off mechanism, i.e. gate value (green) or null-slot mass (orange).

models. The gate stays flat throughout, including when it starts nearly closed. In deep off-by-one models, sinks form first and are then slowly replaced by null-slot mass.

The loss prefers the gate. To ask which route the loss "wants", we add two zero-valued probe scalars per head, active only on non-trigger rows: p_{sink} is added to the bos attention logit and p_{gate} to the gate pre-activation. They leave the forward pass unchanged; their gradients measure how much the loss would decrease per unit logit change along each route. Both act on logits, so they are directly comparable. Figure 5 (single-layer gated model, 3 seeds) shows that during the decisive phase the pull towards closing the gate is larger than the pull towards the sink (peak 6.2×10^{-4} vs. 3.9×10^{-4}). Yet bos mass rises from 0.16 to 1.0 while the gate moves from 0.50 to 0.45. The outcome is therefore decided by how fast each route can move, not by which route the loss prefers.

Explanation. The sink logit on row i is $(W_Q x_i)^\top (W_K x_{\text{bos}})$, a product of two learned factors. As in deep linear networks (Saxe et al., 2014), such products start slowly from a small initialisation and then grow faster the larger they get, which produces the sigmoidal escape seen in Figure 4. The gate pre-activation $W_G x_i + b_G$ is linear in its parameters, and only one input feature (the token-type indicator) separates trigger from non-trigger rows, so with Adam it moves at a roughly constant rate. Once the sink has absorbed the error, the gradient on the gate vanishes and the gate stays where it is. A similar implicit bias of softmax parameterisations towards low-entropy solutions was analysed by Varre et al. (2026).

Intervention: speed up only the gate. If the outcome is a race, giving the gate a head start should change the winner. We multiply the learning rate of the gate parameters (W_G, b_G) by λ and leave everything else unchanged (Figure 6, Table 4). As λ grows, the gate turns into a switch and the sink disappears. At 4 layers, $\lambda = 10$ already removes every sink (0/80 heads, bos mass at the uniform level). It also cuts training time from 16.8k to 3.3k steps,

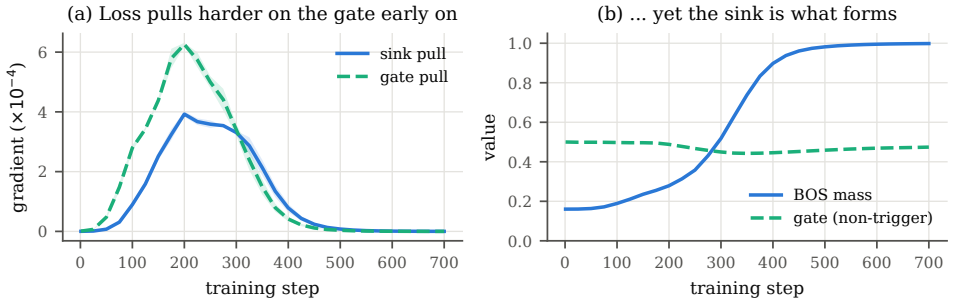


Figure 5: Gradient probes in a 1-layer gated model (mean ± std over 3 seeds). (a) The loss pulls harder on closing the gate than on the sink. (b) The sink nonetheless forms and the gate stays near 0.5.

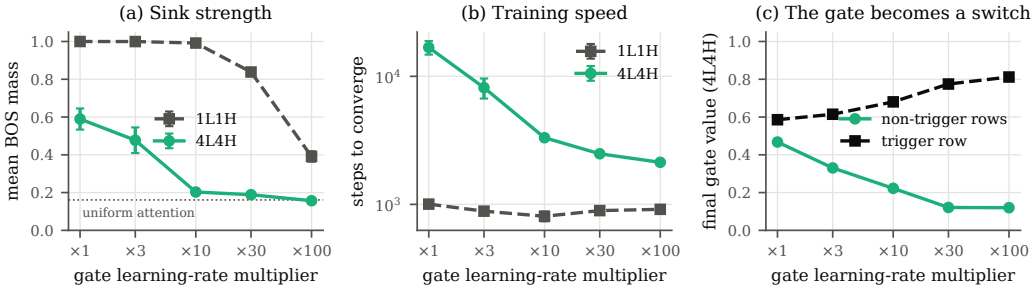


Figure 6: Raising only the gate's learning rate (5 seeds per point). (a) Sinks vanish. (b) Training gets faster. (c) At 4 layers the gate separates trigger from non-trigger rows.

and $\lambda = 100$ cuts it to 2.1k. At 1 layer, $\lambda = 10$ gives a perfect switch and a sink, and the sink weakens only for $\lambda \geq 30$. The sink is thus not merely an alternative solution but a slower one, and training locks into it because it gets there first.

8 Discussion

Implications for the original paper. Ran-Milo (2026) show what softmax needs; we show what training prefers. Adding an off switch makes the sink unnecessary but does not by itself make models sink-free, because gradient descent finds the sink first. Changing how off states are represented is therefore not sufficient on its own; the off mechanism must also be learnable at least as fast as the sink. In our setting a per-parameter-group learning rate is enough to achieve that.

Relation to gated LLMs. Qiu et al. (2025) observe far fewer sinks with gating at scale, while Rizwan et al. (2026) do not at small scale. A race explains both: whether the gate wins depends on the relative speed of the two routes, which is affected by width (the sink logit is a sum over d_{head} products), normalisation layers, initialisation, learning rate and training length. All of these differ between the two studies. Our results suggest checking gate statistics early in training and tuning the gate's learning rate when sink-free behaviour is desired.

Limitations. All experiments use the synthetic task and attention-only models of the original paper: no MLPs, LayerNorm or positional encoding, sequence length 16 and width 16. The constructions are existence results, not a characterisation of all solutions. The speed-race account is supported by probes and an intervention, but we give no formal dynamics theorem. Whether a faster gate removes sinks in language models trained on real text is the

Table 4: Gate learning-rate multiplier λ (gated attention, 5 seeds each, all converged).

	1 layer, 1 head			4 layers, 4 heads		
λ	BOS mass	Gate off/trig.	Steps	Sink heads	Gate off/trig.	Steps
1	1.00	0.50 / 0.78	1004	38/80	0.47 / 0.59	16816
3	1.00	0.33 / 0.93	885	25/80	0.33 / 0.61	8161
10	0.99	0.01 / 0.99	807	0/80	0.22 / 0.68	3315
30	0.84	0.00 / 1.00	894	0/80	0.12 / 0.77	2484
100	0.39	0.00 / 1.00	915	0/80	0.12 / 0.81	2130

most important open question. Some numbers differ slightly between CPU and GPU runs of the same seed, so we report means over 5 seeds throughout.

9 Conclusion

Attention sinks are necessary for softmax, but they are also preferred by training whenever a stable anchor exists, even in architectures that could do without them. In a controlled setting we showed that gated and off-by-one attention admit sink-free solutions, yet gradient descent reliably builds sinks in them because the bilinear sink outruns the gate. Removing the anchor, adding depth, or simply letting the gate learn faster shifts the balance, and in the last case it also speeds up training several-fold.

Code. All code, configurations and raw results are available at https://github.com/REPLACE-WITH-REPO-URL. Every number and figure can be regenerated with reproduce.sh.

References

Yelysei Bondarenko, Markus Nagel, and Tijmen Blankevoort. Quantizable transformers: Removing outliers by helping attention heads do nothing. In Advances in Neural Information Processing Systems (NeurIPS), 2023.

Xiangming Gu, Tianyu Pang, Chao Du, Qian Liu, Fengzhuo Zhang, Cunxiao Du, Ye Wang, and Min Lin. When attention sink emerges in language models: An empirical view. In International Conference on Learning Representations (ICLR), 2025.

Tianyu Guo, Druv Pai, Yu Bai, Jiantao Jiao, Michael I. Jordan, and Song Mei. Active-dormant attention heads: Mechanistically demystifying extreme-token phenomena in LLMs. arXiv preprint arXiv:2410.13835, 2024.

Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In International Conference on Learning Representations (ICLR), 2015.

Evan Miller. Attention is off by one. Blog post, https://www.evanmiller.org/attention-is-off-by-one.html, 2023.

Zihan Qiu et al. Gated attention for large language models: Non-linearity, sparsity, and attention-sink-free. In Advances in Neural Information Processing Systems (NeurIPS), 2025.

Jason Ramapuram et al. Theory, analysis, and best practices for sigmoid self-attention. In International Conference on Learning Representations (ICLR), 2025.

Yuval Ran-Milo. Attention sinks are provably necessary in softmax transformers: Evidence from trigger-conditional tasks. arXiv preprint arXiv:2603.11487, 2026.

Sara Rizwan, Samaanah Abdus Salam, and Mohammed Mudassir Uddin. Do new attention mechanisms actually fix attention sinks at million-token context? arXiv preprint arXiv:2609.08574, 2026.

Andrew M. Saxe, James L. McClelland, and Surya Ganguli. Exact solutions to the nonlinear dynamics of learning in deep linear neural networks. In International Conference on Learning Representations (ICLR), 2014.

Mingjie Sun, Xinlei Chen, J. Zico Kolter, and Zhuang Liu. Massive activations in large language models. In Conference on Language Modeling (COLM), 2024.

Aditya Varre, Mark Rofin, and Nicolas Flammarion. Gradient flow polarizes softmax outputs towards low-entropy solutions. arXiv preprint arXiv:2603.06248, 2026.

Richard Zhe Wang. Abstention and noise filtering: Two missing primitives of softmax attention. arXiv preprint arXiv:2609.22005, 2026.

Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. Efficient streaming language models with attention sinks. In International Conference on Learning Representations (ICLR), 2024.

A Implementation details

Code. We re-implemented the task and models from scratch in PyTorch, following the released reference code of Ran-Milo (2026) for data generation, initialisation, optimiser and stopping rule. Differences from the reference: batched data generation on the device, additional attention variants, probes and tracking. The reference stopping threshold (5×10^{-3}, from the code) is used throughout.

Hyper-parameters. $d = 16$, $L = 16$, batch 128, Adam $(0.9, 0.95)$, no weight decay, initialisation $\mathcal{N}(0, 0.02^2)$ for all weights, gate bias 0 unless stated. Learning rate 10^{-3} for single-layer non-residual models (Table 1 top block, gate-LR 1-layer sweep, probes) and 10^{-4} otherwise. The depth sweep uses residual models with 4 heads for all depths, including $D = 1$. Maximum steps: 50k (80k for 6 and 8 layers); the no-bos 1-layer sweep was capped at 20k steps.

No-bos variant. Position 0 becomes a content token, the trigger is sampled from $\{1, \ldots, 15\}$, and the target averages positions $0..j$. ReLU and sigmoid-n divide by the number of non-bos keys, which is $i + 1$ without bos instead of i.

Probes. Every 25 steps we draw 512 fresh sequences, add zero scalars p_{sink} (to the bos logit) and p_{gate} (to the gate pre-activation, all coordinates) on non-trigger rows of each head, and record $-\partial L/\partial p_{\text{sink}}$ and $\partial L/\partial p_{\text{gate}}$. Probes are removed before the optimisation step, so training is unaffected.

Compute. Single-layer and shallow sweeps ran on a laptop CPU; deeper sweeps ran on Kaggle (2× NVIDIA T4). In total we trained 312 models (plus 3 probe runs); the longest single run (8 layers, on a shared T4) took about 25 minutes.

B Additional results

Sink heads by depth (5 seeds, 4 heads per layer, sink head: bos mass > 0.6):

Layers	1	2	3	4	6	8
softmax	19/20	24/40	29/60	37/80	43/120	62/160
off-by-one	20/20	26/40	36/60	40/80	26/120	7/160
null-slot share in sink heads	0.02	0.02	0.11	0.19	0.32	0.33
gated	20/20	40/40	47/60	38/80	50/120	58/160
gate off / trig.	.42/.66	.43/.62	.45/.60	.47/.59	.47/.57	.48/.56

Gate initialisation (4 layers × 4 heads, 5 seeds): initial gate 0.02: 80/80 sink heads, final gate 0.02/0.05; 0.12: 80/80, 0.11/0.22; 0.50: 38/80, 0.47/0.59; 0.88: 39/80, 0.86/0.90; 0.98: 38/80, 0.98/0.98 (off/trigger).